

Rishabh Shukla

SOFTWARE ENGINEER

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SUMMARY

Backend Software Engineer with **4 years of experience** developing microservices and full-stack applications. Skilled in **Python, Go, Django, FastAPI, React, Docker, Kubernetes, and AWS**, with hands-on experience designing and deploying **100+ high-performance APIs** and **distributed systems**. **AWS and GoLang** certified, with additional expertise in **Machine Learning** for data-driven solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, Go, JavaScript, TypeScript, Java, SQL
- **Frameworks & Libraries:** Django, Flask, FastAPI, Django REST Framework, React, REST APIs, GraphQL, gRPC, Celery, RabbitMQ, Kafka
- **Cloud & DevOps:** AWS (Lambda, EC2, ECS, S3, CloudFront, DynamoDB, SNS, SQS), Azure, Docker, Kubernetes, Terraform, Jenkins
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, NoSQL, Database Design
- **Machine Learning & AI:** scikit-learn, pandas, NumPy, marshmallow, pydantic, SciPy, NLP, Transformers, Computer Vision
- **Version Control & CI/CD:** Git, GitHub, GitLab, Jenkins, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Reliance IQ

Software Engineer, US

Sep 2025 – Jan 2026

- Implemented **scalable REST APIs** in **Python/Flask** to support secure **data ingestion into Firestore**, improving data consistency by **40%** while handling **600 daily requests**.
- Designed a **high-throughput file ingestion pipeline** capable of processing **200K+ row uploads**, **automating validation** and **batching** within a **cross-functional team of 10**.
- Instituted **secure, event-driven** file uploads using **GCS signed URLs**, reducing operational effort by **60%** through time-bound, method-restricted URLs aligned with **security best practices**.
- Refactored **schema validation**, **encoding handling**, and **error isolation logic**, bolstering **backend fault tolerance** and providing stable APIs to unblock **frontend teams**, reducing inconsistencies by **80%**.
- Scheduled **deployment workflows**, shrinking release cycles to under **3 minutes** and improving **deployment reliability** for faster turnaround on backend changes.

Tech Stack: Python, flask, marshmallow, GCP, JavaScript, React, PyTest, Docker, Notion, Github.

Incedo Inc.

Software Engineer, India

Jan 2021 – Dec 2022

- Scaled **distributed backend services** using **Django and Go**, supporting **~1,200,000 monthly API requests**, while **optimizing** service interactions to **reduce infrastructure spend by ~18%** across shared environments.
- Improved system reliability via contract-level and **integration testing (PyTest)**, decreasing production regressions by **35%** and avoiding incident-related costs.
- Optimized **containerized** workloads on **Kubernetes**, collaborating with platform and SRE teams to maintain **99%+ availability** while reducing compute waste during **peak traffic**.
- Strengthened **CI/CD pipelines** with safer rollout strategies and **automated** checks using **Jenkins, Docker, and Helm** reducing **deployment failures by 40%**.
- Authored **API contracts, runbooks, and deployment documentation**, **accelerating onboarding by 30%** and reducing dependency on **senior engineers**, freeing capacity for **higher-impact** system improvements.

Tech Stack: Python, Go, JavaScript, Django, FastAPI, Kubernetes, AWS, Jenkins, GitOps, PyTest, Docker, Terraform, Confluence, Helm, GitHub, GraphQL

Airbnb

Software Engineer, India

Mar 2020 – Dec 2020

- Optimized **backend** request handling across **Django and PostgreSQL** services, sustaining performance for **5K-20K API requests per day** across internal and partner-facing workflows.
- **Reduced peak latency by 30-40% on high-frequency endpoints**, improving **responsiveness** for **downstream consumers** through query tuning and **API-level optimizations**.
- Reduced **backend load by ~60%** by introducing **Redis caching** and **asynchronous** task execution using **Celery**, improving average **response time from ~500ms to ~180ms**.
- Pioneered refined alert thresholds using **Prometheus**, curtailing alert noise by **25%** and improving on-call engineer efficiency alongside **3 key database queries**, improving response times by **20%**.

Tech Stack: Python, Django, JavaScript, PostgreSQL, Go, gRPC, Redis, Celery, Prometheus, Grafana, Jenkins, PyTest, Docker, Terraform, Kubernetes, GitHub, GraphQL

- Improved **data consistency**, cutting operational intervention time by **8–10 hours per week** by reducing reconciliation and manual correction issues.
- **Increased test coverage by ~35%** through **Streamlined API testing with PyTest and Postman**, catching regressions earlier in the release cycle.
- Resolved backend defects, eliminating **3–5 hours of troubleshooting** per release by addressing a **race condition** in the order-processing API.

Tech Stack: Python, Go, Django, JavaScript, MySQL, Postman, PyTest, Jenkins, Git, Docker, Terraform, Kubernetes, GitHub

PROJECTS

Nextgen Bank

- Developed a secure **banking application** in **Django** with **OTP authentication**, reducing unauthorized login attempts by **30-40%**.
- Built core banking features including **fund transfers and real-time account updates**, processing **~5,000 daily transactions**, improving transaction reliability and reducing reconciliation issues.
- Designed and integrated **virtual cards** functionality, enabling secure online payments with **~1,000 virtual cards** generated monthly, improving **user privacy** and reducing fraud exposure.
- Fortified application security with **hashed passwords** and **rate-limiting**, decreasing fraud-related support tickets by **20%**.
- Modernized legacy banking system using **Django and RESTful APIs**, resulting in **~30% improvement in API response times**, enhancing **transaction speed** and **system responsiveness**.

Distributed Microservice Application

- **Architected** and developed **microservices** including **Authentication, Logging, Mail, and Frontend** services using **Go**, **reducing average latency by ~50–70 ms**, improving overall system responsiveness.
- Engineered inter-service communication via **REST APIs, gRPC, and RabbitMQ**, handling **~10–20k requests per second**, ensuring reliable and low-latency data flow between services.
- **Deployed** and managed **microservices** using **Docker Swarm and Kubernetes**, maintaining **99.99% uptime** through **Orchestrated CI/CD pipelines**.

EDUCATION

University of Alabama Birmingham

Birmingham, AL

Master of Science in Computer Science

Aug 2023 – May 2025

Gyan Ganga Institute of Technology & Science

MP, India

Bachelor of Technology in Computer Science and Engineering

Sep 2016 – May 2020

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** – Amazon Web Services | Jun 2021
- **HackerEarth Machine Learning Challenge** – Achieved Global Rank 19 | May 2021
- **Programming with Google’s GoLang Specialization** – Coursera (University of California, Irvine) | Sep 2020
- **TensorFlow Developer Certificate** – Coursera (deeplearning.ai) | Aug 2020